

$$e^{PPP} = \delta; \alpha_0^{PPP} = K - \frac{\delta^2}{2},$$

$$\alpha_1^{PPP} = (1 - \lambda)(b_H - b_L) + K - \frac{\delta^2}{2} + (2\lambda - 1)s$$

This will provide an expected VFM in the PPP case given by:

$$E(\text{VFM}^{PPP}) = (1 - \mu(1 - \lambda))b_H + \mu(1 - \lambda)b_L - K + \frac{\delta^2}{2} - 2\mu\lambda s \quad (6)$$

Then, by setting $t = 1$ and $\lambda = 1$, we can construct the PUB alternative. This will yield:

$$e^{PUB} = 0; \alpha_0^{PUB} = K; \alpha_1^{PUB} = K + s$$

As a result of the lack of profit incentives for F, no effort or innovation will be forthcoming. The fact that it holds all the bargaining power, however, does mean that G will have to pay no more than the costs actually incurred by F. The expected VFM under PUB will be given by:

$$E(\text{VFM}^{PUB}) = \mu(b_H - K - 2s) + (1 - \mu)(b_H - K) = b_H - K - 2\mu s \quad (7)$$

Our second, and key, set of results highlights the intuitive trade-off between efficiency and flexibility in the choice of procurement mode.

Result 2: Under the objective of maximizing VFM, a PPP procurement will dominate public procurement (PUB) when:

$$E(\text{VFM}^{PPP}) - E(\text{VFM}^{PUB}) = \frac{\delta^2}{2} - \mu(1 - \lambda)(b_H - b_L) + 2(1 - \lambda)\mu s > 0 \quad (8)$$

Therefore, procurement using a PPP is more likely to dominate procurement under PUB: the greater is the cost-reducing effect of effort (δ), the smaller is the probability the project design will need to change (μ); the greater is the switching cost (s), and the smaller is the difference between the social values of the best project and the other project ($b_H - b_L$). Assuming that the net benefit of renegotiation ($b_H - b_L - 2s$) is always positive, then the VFM of a PPP is relatively higher when government is in a stronger bargaining position (λ).

The basic trade-off between the efficiency of private providers and the flexibility of public providers comes through very clearly here. When flexibility does not matter because demand is extremely unlikely to change ($\mu = 0$), $E(\text{VFM}^{PPP}) > E(\text{VFM}^{PUB})$, implying that PPP procurement must dominate PUB because of its efficiency benefits. On the other hand, when the need to provide extra incentive for efficiency is less important, perhaps because there is little potential for innovation in the project ($\delta = 0$), procurement via PUB will dominate – that is, $E(\text{VFM}^{PPP}) < E(\text{VFM}^{PUB})$.

Other intuitive results follow from (8) quite simply. When the difference between the “right” project and the “wrong” one ($b_H - b_L$) is large, it is more important for G to

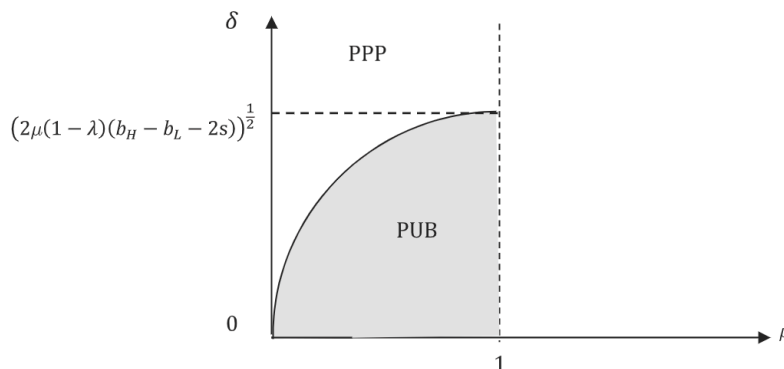
renegotiate – improving the terms that F can extract and reducing the VFM benefits of the PPP. Greater bargaining power on the part of the government expands the range of parameter values under which PPP procurement is preferred by the VFM standard as greater bargaining power on the part of the government limits the ability of the private partner to raise the government's costs in renegotiation. Finally, higher levels of switching costs (s), other things being equal, tend to favor the PPP mode. Under PUB, the public provider bears the full switching costs of both partners. However, under the PPP structure – and given Nash bargaining in renegotiation – the private partners will end up sharing the loss of surplus associated with higher switching costs. Less than 100 per cent of this cost is then passed on to the public sector.

We can illustrate the efficiency–flexibility trade-off with a simple graph based on values of δ (capturing the importance of effort for efficiency) and μ (capturing the need for flexibility) that lead the PPP and PUB procurement modes to deliver identical levels of VFM – i.e. that set condition (8) equal to zero. These levels will be given by $\delta = (2\mu(1 - \lambda)(b_H - b_L - 2s))^{\frac{1}{2}}$, which is illustrated in Figure 1, where the area under the curve (shaded) is the range within which PUB dominates PPP while the area above represents values such that the PPP mode dominates. Other aspects of *Result 2* can be illustrated with this graph. For instance, increases in $b_H - b_L$ or decreases in s rotate the curve upward (around the origin), which expands the area in which PUB dominates.

The interpretation of *Result 2* is therefore quite straightforward. Under a PPP contract, F has incentives to exert cost-reducing effort because it captures the gains of its efforts. Those incentives are absent under PUB. The disadvantages of the PPP, however, derive from the incompleteness of the contract. For the government, this involves new costs – the extra surplus that must be transferred to the private partner under Nash bargaining, which lowers the government's VFM.

Therefore, PPPs will be more attractive when there is little chance for a need to redesign the project – road or bridge projects may come to mind as examples. On the other hand, when it is more likely that the kind of services needed in the future could be very different from those anticipated *ex ante* – as might be the case for sophisticated healthcare projects, for example – the flexibility of public procurement contracts may make them superior.

Figure 1. The efficiency–flexibility trade-off



Extensions

*A Total Social Surplus (TSS) Objective*³²

In the baseline case, the government pursues a value-for-money objective, a measure widely used in the PPP literature and by PPP practitioners. However, this is not the most commonly assumed objective for governments in normative policy work by economists. More typically, government objectives would be modeled as the maximization of some measure of social welfare or total social surplus. This is a different measure than value for money to be sure – principally differing as a result of transfers from one party to another. It has been argued – for example, by public sector labor unions – that PPPs facilitate the substitution of poorly paid private sector labor for more highly paid (and unionized) public sector labor. To the extent that this happens, these transfers do improve value for money, but do not represent increases in total social surplus. Profits earned by private providers, on the other hand, would harm value for money but not necessarily affect total social surplus.

To study this alternative objective, however, we need to add a deadweight cost of government financing (from tax revenues), $\gamma > 0$, which represents the additional cost to the economy when a government extracts \$1 to pay for the project. This is because without some shadow price of public funds that incorporates the cost of raising revenue, there will not be a unique solution to the question of what price the government will pay for the services provided.

To facilitate comparison with VFM analysis, we continue to focus on the two special cases: (i) PUB with $\lambda = 1$, $t = 1$, and (ii) PPP with $0 \leq \lambda \leq 1$, $t = 0$. By applying similar analysis as in the VFM case above but adding in this cost of government financing, we get:

$$E(TSS^{PUB}) = b_H - (1 + \gamma)K - 2(1 + \gamma)\mu s$$

$$E(TSS^{PPP}) = b_H + (1 - \lambda)b_L - (1 + \gamma)(K - \frac{\delta^2}{2}) - 2(1 + \gamma)\lambda\mu s$$

Comparison of $E(TSS^{PUB})$ and $E(TSS^{PPP})$ leads to a set of results similar to those in Result 2. A new result relates to the added parameter, the marginal deadweight loss of government funding (γ). It is straightforward to show that increases in γ favour the PPP alternative. This is because increases in the deadweight loss of government funding will favour the procurement mode that involves the lowest cost to government. The PPP mode lowers the cost of production because of the higher levels of effort, and it also shares the switching costs with the private partner whose financing does not create deadweight loss.

Would a government pursuing a VFM objective adopt the PPP mode when the PUB approach would yield greater total surplus under the TSS objective? Comparing the TSS and VFM results provides our third set of results, which demonstrate that the different objectives can indeed lead to different choices.

Result 3: When the public and private partners have different bargaining weights the following cases become possible:

- (i) When the government has the greater bargaining weight (i.e. $\lambda > 1/2$) it is possible for a PPP to maximize VFM while PUB maximizes TSS;

- (ii) When the government has the lesser bargaining weight (i.e. $\lambda < 1/2$) it is possible for a PPP to maximize TSS while PUB maximizes VFM.
- (iii) When the government and firm have equal bargaining weight (i.e. $\lambda = 1/2$), comparisons under VFM are the same as those under TSS. They are also the same if the deadweight cost of government financing (γ), the probability of change (μ) and/or the switching costs (s) are zero.

Some intuition for this result comes from considering the case in which the government has a limited amount of bargaining power (λ is close to 0). In such cases, the government will pay a high price for any changes it makes, pushing down the VFM, but not necessarily damaging TSS much since most of the high renegotiated prices are simply transfers.

Toll Revenue PPPs

In the PPP model we have used to this point, the private partner is paid a fixed sum to deliver the project. This could have been in the form of a lump sum or a stream of “availability payments” (i.e. payments dependent only on the facility operating but not on the actual demand for that facility or the extent of its usage). In many PPP arrangements, however, private parties are paid according to the use of the services. Road projects funded by tolls are a common example in which users rather than the government pay the private partners. In other cases private partners are paid according to demand or use of the facilities, but are paid by the government.³³

In cases in which the private partner is paid based on the project’s success in terms of meeting demand, we can imagine that this player will be much more willing – even eager – to amend the project if demand changes make design changes optimal. Similarly, users (or the government if it pays) are not as disadvantaged by a failure to adapt to change since they will not have to pay as much. Put another way, the players’ threat points are different in such a situation.

In exploring a toll-based PPP alternative, we will continue to assume that $t = 0$ in the PPP mode with tolls. Suppose that, under the PPP contract, the fraction of the benefit that F gets (via usage fees) is τ_0 . The government’s VFM objective here then will be to secure the best project by surrendering the smallest fraction of the benefits to the private partner.

Given τ_0 , F chooses an optimal level of effort to maximize its profit. Once again F ’s incentive compatibility constraint is:

$$\text{Max } \pi = \text{Max } \tau_0 b_H - (K - \delta e) - \frac{e^2}{2}$$

and the optimal effort level is again $e^{PPP} = \delta$.³⁴

Potential private partners will bid according to what level of τ_0 they are willing to accept, with the contract going to the lowest bidder. We maintain the assumption that the winning firm, F , will have bid to the point where it will break even (its limited liability constraint) without renegotiation. This implies:

$$\pi = \tau_0 b_H - (K - \delta e^{PPP}) - \frac{(e^{PPP})^2}{2} = \tau_0 b_H - K + \frac{\delta^2}{2} = 0$$

$$\text{so } \tau_0^* = \frac{1}{b_H} \left(K - \frac{\delta^2}{2} \right)$$

If changes in demand materialize, G and F renegotiate and sign a new contract. G's threat point is $(1 - \tau_0^*)b_L$. If we think of $\tau_0^*b_L$ as G's payment to F, then G actually "pays" less now as $\tau_0^*b_L < \tau_0^*b_H$. In this sense, we say that G has a stronger bargaining position than it would in an availability case. If the firm does not agree to change the contract, its payoff is $\tau_0^*b_L - K + \frac{\delta^2}{2}$, which is negative, as $b_L < b_H$. Consistent with our earlier assumption, we assume that F can walk away. So in this case F's threat point still generates zero profits (in this case through exit). Assume that the target of G is to maximize VFM. The Nash Product is then:

$$NP = [(1 - \tau_1)b_H - s - (1 - \tau_0^*)b_L]^\lambda [\tau_1 b_H - K + \frac{\delta^2}{2} - s - 0]^{1-\lambda}$$

The new toll rate, τ_1^* , will then be given by:

$$\tau_1^* = \frac{1}{b_H} \left((1 - \lambda)(b_H - b_L) + (\lambda b_H + (1 - \lambda)b_L) \frac{1}{b_H} \left(K - \frac{\delta^2}{2} \right) - (1 - 2\lambda)s \right)$$

Therefore, the expected VFM of the toll contract is:

$$\begin{aligned} E(\text{VFM}_{\text{toll}}^{\text{PPP}}) &= \mu((1 - \tau_1^*)b_H - s) + (1 - \mu)(1 - \tau_0^*)b_H \\ &= ((1 - \mu(1 - \lambda))b_H + \mu(1 - \lambda)b_L)(1 - \tau_0^*) - 2\lambda\mu s \\ &= ((1 - \mu(1 - \lambda))b_H + \mu(1 - \lambda)b_L) \left(1 - \frac{1}{b_0} \left(K - \frac{\delta^2}{2} \right) \right) - 2\lambda\mu s \end{aligned}$$

We can now ask whether a toll contract or an availability contract generates greater VFM for the government. Recall that expected VFM of the availability contract, for clarity here now labelled $E(\text{VFM}_{\text{avail}}^{\text{PPP}})$, is given by (6).

This leads to our fourth set of results.

Result 4: When the objective of the government is to maximize VFM, the toll contract dominates the availability contract. (Proof provided in the online Appendix.)

Thus, the toll contract dominates the availability contract in terms of VFM. Additionally, we see that the advantage of the toll contract over the availability contract is greater: the more likely it is that the project will need to change (higher μ), the greater the cost of the project (bigger K), the smaller the cost-reducing effect of effort (δ) and the greater the percentage difference between the "right" and "wrong" project (larger $(b_H - b_L)/b_H$). This is because under the toll contract, the government essentially pays less if renegotiation fails and hence the government is in a stronger bargaining position. Finally, as λ gets larger, renegotiation is less costly to G, with the result that the VFM

advantages of the toll contract are greater when G has less bargaining power (i.e. when λ is small).

It follows trivially, then, that when tolling is to be used to pay for the project, the PPP model will dominate traditional public provision under an even broader set of conditions than would an availability payments-based PPP.³⁵

Conclusions and Directions for Future Research

Public–private partnerships are being used to deliver public services in many sectors, such as transportation, water, healthcare, education and prisons. In this paper, we have examined an important trade-off associated with the choice between PPP and more traditional public procurement methods (PUB). While the PPP model provides the private contractor with greater incentives for cost-reducing effort and innovation, it locks the government into a long-term contract that may be costly to renegotiate if changing circumstances make a project redesign optimal. We show that the PPP model will be superior when possible efficiencies are large, the probability there will be a need to change the project is small, the gains through project redesign are small and when the government’s bargaining power in renegotiation is greater. This result holds whether government’s objective is to maximize value for money (VFM) or total social surplus (TSS), though the different objectives can imply different choices between the PPP and PUB approaches. Our analysis has also demonstrated that PPP contracts based on usage-sensitive payments (e.g. tolls) can generate higher VFM for the government.

To our knowledge, this paper is the first to offer a formal model that focuses on this trade-off, though the idea that efficiency comes at the price of flexibility is very intuitive and appears in other contexts (e.g. in the choice of flexible vs inflexible production technologies). Our results may shed some light on why PPPs have become popular in some sectors such as roads and water – where it could be argued that the need for large changes to designs is likely to be relatively smaller – while they remain less common (and more controversial) in areas that might seem more dynamic, such as healthcare and information technology.

A key issue for further work relates to some of the strategies that public and private sector partners employ to deal with the deficiencies of the PPP and PUB models presented above. For example, in PPP projects the parties may anticipate the need to renegotiate in the future and may then put into place mechanisms (e.g. third-party arbitration) to limit the ability of the private partner to take advantage of its strong position to extract much higher payments. A particularly promising innovation directed at reducing the costs of adapting to changing economic (particularly, but not exclusively, demand-related) conditions is the “Least-Present-Value-of-Revenue (LPVR) auctions” of Engel et al. (2001). Under this mechanism, which has been applied in South American projects, private parties bid on the lowest net present value of revenues they will accept to undertake the project. The winning bidder then holds the contract, collecting tolls or other payments, until this net present value threshold is reached. It is easy to see how this mechanism can help avoid costly renegotiation due to demand shocks, for example. If demand is greater than expected, which might otherwise have created great public pressure to prevent a private windfall, the concession period is automatically shortened without any further negotiations necessary. Similarly, if demand is lower than anticipated, rather than threatening to

shut down to avoid losses on an existing contract, the private partner will simply see its concession extended to allow it to recover its sunk investments over a longer period.

PUB modes can also be improved by creating incentives for innovation and effort on the part of public sector providers. Indeed, a great deal of effort has been put into making governments more efficient providers of services generally – for example, by giving managers more authority for the way their units operate but also making them more accountable for the quantity and quality of the services provided.

Finally, though it would require a model different from ours here, it would be worthwhile to explore some of the potential benefits of the relative inflexibility of PPP contracts. The fact that these arrangements are more difficult to change may actually yield benefits of at least two types: (i) they compel governments to make decisions based on efficient life-cycle cost analyses rather than building assets today and worrying about operations and maintenance costs later; and (ii) they may provide a commitment vehicle which assures that critical maintenance is not deferred – a very common problem with public infrastructure assets.³⁶

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Supplemental Data

The Appendix for this article can be accessed at <http://dx.doi.org/10.1080/13876988.2015.1029333>

Notes

1. An early version of PPPs was the Private Finance Initiative (PFI) pioneered in the UK from the early 1990s.
2. Organization for Economic Cooperation and Development (OECD) (2008). For an interesting discussion of the wide range of PPP models and some of the history behind public–private cooperation in delivering public services, see Hodge and Greve (2013) and the references cited therein.
3. The private sector partner is typically a consortium of firms acting through a special purpose vehicle to deliver the assets and associated services. One major way PPPs differ from one another is according to which of the tasks are assumed by the private partner. For example, in the common “DBFOM” form the private partner “designs, builds, finances, operates and maintains” the assets but in other cases the private partner will assume a smaller set of tasks.
4. Infrastructure Journal Project Database, <http://www1.ijonline.com> (accessed July 25, 2011).
5. See, e.g., De Bettignies and Ross (2004) for a discussion of what distinguishes modern PPPs from classical contracting out.
6. There has not been enough work done to properly evaluate the success of PPP projects *ex post*, in part because these are generally very long-lived – and on-going – agreements, and a full accounting cannot really be done until the agreements have expired. The UK was an early adopter of the model and work by the National Audit Office there suggests some success with the private finance initiative (PFI) version of the PPP model – there are many NAO reports at <http://www.nao.org.uk/>. Australia has also been a leader in the

use of the PPP model, and there is evidence of success there as well, see e.g. Infrastructure Partnerships Australia (2007).

7. For example, see Benz et al. (2002), Hart (2003), Hart et al. (1997), Bennett and Iossa (2006) and Iossa and Martimort (2015). For a discussion of other aspects of PPPs studied by economists, see De Bettignies and Ross (2011).
8. Not surprisingly, it is this aspect of PPPs that led to them being strongly opposed by public sector unions. For example, see “The case against PFI” from the website of the largest public service union in the UK (UNISON): <http://www.unison.org.uk/pfi/caseagainst.asp>.
9. In the UK, PPP (or PFI) is not recommended for the provision of information technology services, in part because of the high probabilities and costs of changes (see Yescombe 2007, p. 27). From HM Treasury (2006, p. 32): “the PFI procurement structure is unlikely to deliver value for money ... where authorities require a significant degree of short-term flexibility due to fast-changing service requirements”.
10. Making the point that the surrendering of decision-making authority to the private sector in PPPs makes it more difficult for the public sector to adapt to changing demands for public services, see e.g. Organization for Economic Cooperation and Development (OECD) (2008, pp. 65–69), Yescombe (2007, section 2.12) and PricewaterhouseCoopers (2005, Chapter 2).
11. Excluding concessions in the telecommunications sector “because practically all telecommunication projects were privatized rather than concessioned, raises the incidence of renegotiation to 41.5%” (Guasch 2004, p. 12). Guasch studied various aspects of these renegotiations, including which party initiated the renegotiation, the form of bidding that resulted in the concession award, the incidence of renegotiation in different sectors, and the average time before contracts are renegotiated. In their study of 61 PPP road concession contracts in Chile, Colombia and Peru between 1993 and 2010, Biltran et al. (2013) find that 50 of the contracts had been renegotiated at least once – and in total there were 540 renegotiations on these contracts. In 80 per cent of the cases the renegotiations were triggered by the government with higher costs then imposed on the public sector.
12. From page 8 of the NAO report: “Under PFI, almost any requested change, even as small as a new electrical socket, has to be processed through the SPV as it manages the asset during the contractual period and bears the risk of failing to meet service obligations. Often lacking the option of going to a different supplier, even for major changes, there is a risk that the public sector will have reduced leverage in negotiation and that the SPV or FM provider may not be incentivised to keep down the cost of changes or to process them quickly.” This report goes on to list the kinds of changes that come up frequently in long-term PPP contracts.
13. Two papers consider flexibility issues quite different from those studied here. Iossa and Martimort (2015) consider the advantages of negotiating sequentially (i.e. with more flexibility) with a builder and operator versus bundling the two tasks. Athias and Saussier (2010) look at the optimal degree of pricing flexibility within PPP contracts.
14. We will highlight the differences between this paper and the analysis of BT at various points below.
15. See also Vining et al. (2005). Going back further, there are clear parallels as well with the “administered contracts” work of Goldberg (1976), who recognized that large value, long-term contracts will almost certainly be incomplete, opening them up to renegotiation for any of a variety of reasons. In some cases this may need to be addressed through some form of regulatory mechanism.
16. In BT’s private sector procurement model, they consider fixed price contracts which have properties similar to our PPP contract, and cost-plus contracts which have incentive properties similar to our PUB arrangements.
17. It is possible that effort is observable. Here we only assume that effort is not contractible, i.e. that its value cannot be demonstrated for a court or any contract enforcement mechanism.
18. In fact, the most interesting cases involve changes in the nature of demand – i.e. the kinds of services to be provided. A bridge redesigned to accommodate more bicycles and fewer cars might be an example. Uncertainty about simply the level of demand for a project can often be dealt with in carefully drafted contracts.
19. In contrast, BT use a “take-it-or-leave-it” renegotiation model.
20. If this is solely a renegotiation cost, it may seem likely that s would be smaller (or even zero) when the changes can simply be ordered by the government under a traditional public procurement process. In such a case it is not difficult to show that higher renegotiation costs (that do not apply under PUB) will make PUB a relatively more attractive option.

21. It is not difficult to show that, because effort affects only costs and not benefits in our model, an alternative timing in which effort decisions are made before changes and renegotiations would produce identical results to those derived here.
22. This requires that the total benefits of making the change ($b_H - b_L$) exceed the costs of making the change ($2s$).
23. A common concern raised about PPPs, particularly when the public partner is an under-resourced government department of a developing country, is that the private partners will have more legal and technical “firepower” at the table than will the government.
24. This is not to deny that there can be a problem of firms adopting strategies in which they win the contract with an apparently attractive bid, only to threaten later that the deal must be renegotiated or they will not continue. However, sophisticated governments will be alert to this possibility, and will want to scrutinize bids carefully to assure themselves that they are feasible. Also, in a world in which most of these private sector PPP players wish to continue to win future bids with the same or other governments, such gaming could be damaging to their reputations as trustworthy partners.
25. The classic reference is Sappington (1983). See also the text by Laffont and Martimort (2002). It can be shown, as well, that the key trade-off between efficiency and flexibility remains even if firms are permitted to bid down to zero *ex ante* expected profits (and can commit to their bids) as long as there are costs associated with renegotiation.
26. If the limited liability constraint is satisfied it is very easy to see that a participation or individual rationality constraint (requiring firms to earn non-negative profits in expectation *ex ante*) will never be binding. For this reason we do not include participation constraints here.
27. As suggested, this does require that the winning bidder can credibly convince the government that its bid will allow it to break even in the absence of any renegotiations.
28. All proofs are contained in the online-only Appendix.
29. If we allow t to exactly equal 1 we lose uniqueness to our solutions – since the government taxes back every dollar of profit, it does not care what price it pays. Therefore, when we speak of $t = 1$ here, we more precisely mean $t = 1 - \varepsilon$, where ε can be an arbitrarily small but positive quantity.
30. An alternative modeling technique could simply involve assuming that under PUB, the government does not need to renegotiate changes and can just order them. This approach, taken in an earlier version of this paper, produces essentially identical results.
31. To be clear, we do not see λ as a choice variable except to the extent that a government can choose to procure using traditional public methods, thereby effectively setting λ to one. Any form of private provision will involve, for the purposes of our model, a bargaining weight that is exogenous to the government.
32. Details on some of the steps not fully reported in this subsection are contained in the online Appendix.
33. Shadow tolls (where use is measured but the tolls are paid by the government) on road and bridge projects would be an example.
34. The optimal effort is the same here in part because effort does not affect demand in this model. It would be interesting to explore the implications of having effort influence demand as well as costs.
35. This is also shown in the online Appendix.
36. We are grateful to an anonymous referee and editor for stressing these important points to us. This referee also suggested a number of interesting supply-side issues worth exploring in future work, including the role of infrastructure pricing (e.g. tolling) on the supply response in PPP arrangements and how PPP arrangements affect the speed of that supply response.

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